ELEC4631 Lecture 8: State-Space Transformations and Pole Placement

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State-space realisations/representations

- Given a transfer function $H(s)$, a state space system $S = (A,B,C,D)$ is said to be a *realisation* or *representation* of $H(s)$ if the transfer function of S is $H(s)$. That is, $C(sI-A)^{-1}B + D = H(s)$.
- Example:

$$S = \left(\begin{bmatrix} 1 & 2 \\ -3 & 0 \end{bmatrix}, \begin{bmatrix} 4 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 & 2 \end{bmatrix}, -2 \right)$$

is a representation of the transfer function $H(s) = (-2s^2+16s-32)/(s^2-s+6)$ since the transfer function of S is $H(s)$.

State-space transformations

- An order n system $S = (A,B,C,D)$ can be transformed to another order n system $S' = (TAT^{-1},TB,CT^{-1},D)$ by a real invertible nxn matrix T.
- If $x(t)$ state of S then state of S' is $z(t) = Tx(t)$.

$$\dot{x}(t) = A\,x(t) + Bu(t);\; x(0) = x_0$$
$$y(t) = Cx(t) + Du(t)$$

$$\Updownarrow$$

$$\dot{z}(t) = TAT^{-1}z(t) + TBu(t);\; z(0) = z_0 = Tx_0$$
$$y(t) = CT^{-1}z(t) + Du(t)$$

State-space transformations

- Transformation from $S = (A,B,C,D)$ to $S' = (TAT^{-1},TB,CT^{-1},D)$ by T called *state-space similarity transformation*.
- In general, two systems $S_1 = (A_1,B_1,C_1,D_1)$ and $S_2 = (A_2,B_2,C_2,D_2)$ said to be *similar* or *equivalent* to one another if $D_1 = D_2$ and there exists an invertible matrix T such that $A_2 = TA_1T^{-1}$, $B_2 = TB_1$ and $C_2 = C_1T^{-1}$.

State-space transformations

- Two equivalent state space systems S_1 and S_2 have same transfer function:

$$\begin{aligned}
H_2(s) &= C_2(sI - A_2)^{-1}B_2 + D_2 \\
&= C_1T^{-1}(sI - TA_1T^{-1})^{-1}TB_1 + D_1 \\
&= C_1T^{-1}(sTT^{-1} - TA_1T^{-1})^{-1}TB_1 + D_1 \\
&= C_1T^{-1}T(sI - A_1)^{-1}T^{-1}TB_1 + D_1 \\
&= C_1(sI - A_1)^{-1}B_1 + D_1 \\
&= H_1(s).
\end{aligned}$$

State-space transformations

- However, converse not true. State-space realisation of a given transfer function not unique and there are unequivalent realisations.
- Example:

$$S_1 = \left(\begin{bmatrix} -1 & 0 \\ -1 & 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \end{bmatrix}, 1 \right)$$

$$S_2 = \left(\begin{bmatrix} -1 & 0 \\ 0.1 & -2 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \end{bmatrix}, 1 \right)$$

are two unequivalent second order realisations of transfer function $H(s) = (s+2)/(s+1)$.

Transformation to control or observer canonical form

- Given some state-space representation S, is there a transformation T that will put it to control or observer canonical form?
- In general, this is not always possible. Representation S needs to have some specific properties.

Transformation to control or observer canonical forms

- Theorem: A SISO LTI system $S = (A,B,C,D)$ has a similarity transformation to control (respectively, observer) canonical form if and only if $\mathscr{C}$ (respectively, $\mathscr{O}$) is invertible.
- That is, a system can be transformed to control (respectively, observer) canonical form if and only if system is controllable (respectively, observable).
- Control canonical form is always controllable, observer canonical form always observable.

Transformation to control canonical form

- If $S = (A,B,C,D)$ has invertible controllability matrix $\mathscr{C}$ and $c_A(s) = \det(sI\text{-}A) = a_0 + a_1 s + \ldots + a_{n\text{-}1}s^{n\text{-}1} + s^n$ then a similarity transformation T that brings S to control canonical form is:

$$T = \begin{bmatrix} 0 & 0 & \cdots & 0 & 1 \\ 0 & 0 & \cdots & 1 & p_1 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 1 & \cdots & p_{n-3} & p_{n-2} \\ 1 & p_1 & \cdots & p_{n-2} & p_{n-1} \end{bmatrix} \mathscr{C}^{-1}$$

where

$$\begin{aligned} p_1 &= -a_{n-1} \\ p_2 &= -a_{n-2} - a_{n-1}p_1 \\ p_3 &= -a_{n-3} - a_{n-2}p_1 - a_{n-1}p_2 \\ &\vdots \\ p_{n-1} &= -a_1 - a_2 p_1 - \ldots - a_{n-1}p_{n-2} \end{aligned}$$

Keep in mind here the definition of characteristic polynomial of A:

$c_A(s) = \det(sI\text{-}A)$

Transformation to observer canonical form

- In similar fashion, if $S = (A,B,C,D)$ has invertible observability matrix $\mathcal{O}$ and $c_A(s) = \det(sI\text{-}A) = a_0 + a_1 s + \ldots + a_{n\text{-}1}s^{n\text{-}1} + s^n$ then a similarity transformation T that brings S to observer canonical form is:

$$T = \begin{bmatrix} 0 & 0 & \ldots & 0 & 1 \\ 0 & 0 & \ldots & 1 & p_1 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 1 & \ldots & p_{n-3} & p_{n-2} \\ 1 & p_1 & \ldots & p_{n-2} & p_{n-1} \end{bmatrix}^{-1} \mathcal{O}$$

where

$$\begin{aligned} p_1 &= -a_{n-1} \\ p_2 &= -a_{n-2} - a_{n-1}p_1 \\ p_3 &= -a_{n-3} - a_{n-2}p_1 - a_{n-1}p_2 \\ &\vdots \\ p_{n-1} &= -a_1 - a_2 p_1 - \ldots - a_{n-1}p_{n-2} \end{aligned}$$

Again, keep in mind the definition of characteristic polynomial of A:
$c_A(s) = \det(sI\text{-}A)$

Example transformations

- Consider system $S = (A,B,C,D)$ with

$$A = \begin{bmatrix} 7 & -2 \\ 3 & 1 \end{bmatrix} \quad B = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad C = \begin{bmatrix} 1 & -2 \end{bmatrix} \quad D = 1$$

Characteristic polynomial of A is $c_A(s) = s^2\text{-}8s+13$, and controllability and observability matrices are

$$\mathscr{C} = \begin{bmatrix} 1 & 5 \\ 1 & 4 \end{bmatrix} \quad \mathscr{O} = \begin{bmatrix} 1 & -2 \\ 1 & -4 \end{bmatrix}$$

and are both invertible. Let T_1 and T_2 be

$$T_1 = \begin{bmatrix} 0 & 1 \\ 1 & 8 \end{bmatrix} \mathscr{C}^{-1} = \begin{bmatrix} 1 & -1 \\ 4 & -3 \end{bmatrix} \quad T_2 = \begin{bmatrix} 0 & 1 \\ 1 & 8 \end{bmatrix}^{-1} \mathscr{O} = \begin{bmatrix} -7 & 12 \\ 1 & -2 \end{bmatrix}$$

Example transformations

- Then

$$S_1 = (T_1 A T_1^{-1}, T_1 B, C T_1^{-1}, D) = \left(\begin{bmatrix} 0 & 1 \\ -13 & 8 \end{bmatrix} , \begin{bmatrix} 0 \\ 1 \end{bmatrix} , \begin{bmatrix} 5 & -1 \end{bmatrix} , 1 \right)$$

Control canonical form

$$S_2 = (T_2 A T_2^{-1}, T_2 B, C T_2^{-1}, D) = \left(\begin{bmatrix} 0 & -13 \\ 1 & 8 \end{bmatrix} , \begin{bmatrix} 5 \\ -1 \end{bmatrix} , \begin{bmatrix} 0 & 1 \end{bmatrix} , 1 \right)$$

Observer canonical form

Transformation to modal canonical form

- Suppose that n x n matrix A has only real eigenvalues and is diagonalisable.
- Then there is a real matrix V that diagonalises A

$$V^{-1}AV = \mathrm{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n) = \begin{bmatrix} \lambda_1 & 0 & \ldots & 0 \\ 0 & \lambda_2 & \ldots & \vdots \\ \vdots & \vdots & \ddots & 0 \\ 0 & 0 & \ldots & \lambda_n \end{bmatrix}$$

V diagonalizes *A*

Transformation to modal canonical form

- Therefore $T = V^{-1}$ is a transformation that brings $S = (A,B,C,D)$ to modal canonical form $S_{\text{modal}} = (TAT^{-1}, TB, CT^{-1}, D)$.
- Example: Consider second order system

$$S = \left(\begin{bmatrix} 1 & -6 \\ 1 & -4 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 & 2 \end{bmatrix}, 1 \right)$$

The matrix $A = \begin{bmatrix} 1 & -6 \\ 1 & -4 \end{bmatrix}$ has eigenvalues -1 and -2, and

diagonalised by

$$V = \begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}$$

Transformation to modal canonical form

- So,

$$T = V^{-1} = \begin{bmatrix} 1 & -2 \\ -1 & 3 \end{bmatrix}$$

and modal canonical form is

$$S_{\text{modal}} = \left(TAT^{-1}, TB, CT^{-1}, D\right)$$
$$= \left(\begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix}, \begin{bmatrix} -1 \\ 2 \end{bmatrix}, \begin{bmatrix} -1 & 0 \end{bmatrix}, 1\right)$$

Control of LTI state-space systems

- If state $x(t)$ is accessible then it is possible to employ state feedback control $u(t) = \eta(x(t))$ for some function η of the state. This is known as *state feedback*.
- We now consider a special kind of state feedback, linear state feedback where η is a linear function and thus $u(t) = -Kx(\mathrm{t})$ for some row vector K of the appropriate length. (Minus sign is for notational convenience and to indicate negative feedback.)

Linear state feedback

- Substituting $u(t) = - Kx(\mathrm{t})$ we get

$$\begin{aligned} \dot{x}(t) &= Ax(t) + Bu(t) \\ &= Ax(t) - BKx(t) \\ &= (A - BK)x(t) \end{aligned}$$

Therefore under this feedback A changes to $A - BK$.

Linear state feedback

- Suppose that $K = [\ K_0\ K_1\ \ldots\ \ K_{n\text{-}1}\ ]$ and system is in control canonical form then A changes to $A\text{-}BK$ as follows

$$A = \begin{bmatrix} 0 & 1 & 0 & \ldots & 0 \\ 0 & 0 & 1 & \ldots & 0 \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & 0 & 0 & \ldots & 1 \\ \boxed{-a_0 & -a_1 & -a_2 & \ldots & -a_{n-1}} \end{bmatrix}$$

$$A - BK = \begin{bmatrix} 0 & 1 & 0 & \ldots & 0 \\ 0 & 0 & 1 & \ldots & 0 \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & 0 & 0 & \ldots & 1 \\ -a_0 & -a_1 & -a_2 & \ldots & -a_{n-1} \end{bmatrix} - \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix} \begin{bmatrix} K_0 & K_1 & \ldots & K_{n-1} \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 1 & 0 & \ldots & 0 \\ 0 & 0 & 1 & \ldots & 0 \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & 0 & 0 & \ldots & 1 \\ -a_0 - K_0 & -a_1 - K_1 & -a_2 - K_2 & \ldots & -a_{n-1} - K_{n-1} \end{bmatrix}$$

Linear state feedback

- In changing system matrix A to A-BK with linear state feedback we also change system's characteristic polynomial from

$$c_A(s) = \det(sI - A) = a_0 + a_1 s + \ldots + a_{n-1} s^{n-1} + s^n$$

to

$$c_{A-BK}(s) = \det(sI - (A - BK))$$
$$= (a_0 + K_0) + (a_1 + K_1)s + \ldots + (a_{n-1} + K_{n-1})s^{n-1} + s^n$$

- Moreover, notice the freedom to make the new characteristic polynomial into anything we desire by appropriately choosing the values of K_0, K_1, …, K_{n-1}.
- Why is this important?

Linear state feedback

- Remember that transfer function $H(s)$ of the system is

$$H(s) = C(sI - A)^{-1}B + D$$

therefore any pole of $H(s)$ must be an eigenvalue of A.

- Thus in control canonical form with linear feedback - $Kx(t)$ we have complete control over where poles will go. This is known as *pole placement*.
- Poles play a central role in determining the dynamic characteristics of system. Ease of controlling them is one reason for the name *control canonical form*.

Linear state feedback

- What if system not in control canonical form?
- Recall from earlier: if system controllable then it can be transformed to control canonical form.
- So, suppose that system controllable and look at what happens.

Linear state feedback

- Suppose that $S = (A,B,C,D)$ controllable and let T transform S into control canonical form.
- With linear state feedback $u(t) = - Kx(\mathrm{t})$ we have

$$\dot{x}(t) = Ax(t) - BKx(t)$$

$$\Big\Downarrow\, T$$

$$\dot{z}(t) = TAT^{-1}z(t) - TBKT^{-1}z(t) \qquad z(t) = Tx(t)$$

$$= (TAT^{-1} - TBK_1)z(t) \qquad K_1 = KT^{-1}$$

Linear state feedback

- Let characteristic polynomial of A and TAT^{-1} (they are the same) be

$$c_A(s) = a_0 + a_1 s + \ldots + a_{n-1} s^{n-1} + s^n$$

and we want to change this to, say,

$$R(s) = r_0 + r_1 s + \ldots + r_{n-1} s^{n-1} + s^n$$

- Since TAT^{-1} and TB in control canonical form what we need to achieve this is to set K_1 to:

$$K_1 = \begin{bmatrix} r_0 - a_0 & r_1 - a_1 & \ldots & r_{n-1} - a_{n-1} \end{bmatrix}$$

Linear state feedback

- Since $K_1 = KT^{-1}$, and (from previously)

$$T = \begin{bmatrix} 0 & 0 & \dots & 0 & 1 \\ 0 & 0 & \dots & 1 & p_1 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 1 & \dots & p_{n-3} & p_{n-2} \\ 1 & p_1 & \dots & p_{n-2} & p_{n-1} \end{bmatrix} \mathscr{C}^{-1}$$

$$\begin{aligned} p_1 &= -a_{n-1} \\ p_2 &= -a_{n-2} - a_{n-1}p_1 \\ p_3 &= -a_{n-3} - a_{n-2}p_1 - a_{n-1}p_2 \\ &\vdots \\ p_{n-1} &= -a_1 - a_2 p_1 - \ldots - a_{n-1}p_{n-2} \end{aligned}$$

we have that

$$\boxed{K = \begin{bmatrix} r_0 - a_0 & r_1 - a_1 & \dots & r_{n-1} - a_{n-1} \end{bmatrix} \begin{bmatrix} 0 & 0 & \dots & 0 & 1 \\ 0 & 0 & \dots & 1 & p_1 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 1 & \dots & p_{n-3} & p_{n-2} \\ 1 & p_1 & \dots & p_{n-2} & p_{n-1} \end{bmatrix} \mathscr{C}^{-1}}$$

Linear state feedback

- Let's write formula for K in more compact form. To do this we need what's called the Cayley-Hamilton Theorem.
- Cayley-Hamilton Theorem: Let A be a n x n square matrix. Let $c_A(s) = \lambda_0 + \lambda_1 s + \ldots + \lambda_{n-1}s^{n-1} + s^n$ be the characteristic polynomial of A, *i.e.*, $c_A(s) = \det(sI\text{-}A)$. Then $c_A(A) = \lambda_0 I + \lambda_1 A + \ldots + \lambda_{n-1}A^{n-1} + A^n = 0$.
- In a nutshell: A satisfies the same characteristic equation as its eigenvalues, with the scalar s replaced with the matrix A and the constant 1 replaced by an identity matrix of a compatible dimension.

Linear state feedback

- Let t_1 be the first row of T (= last row of $\mathscr{C}^{-1}$), then it can be shown that (tutorial exercise)

$$T = \begin{bmatrix} t_1 \\ t_1 A \\ \vdots \\ t_1 A^{n-2} \\ t_1 A^{n-1} \end{bmatrix}$$

so we can write

$$K = t_1 \sum_{k=0}^{n-1} r_k A^k - t_1 \sum_{k=0}^{n-1} a_k A^k$$

Linear state feedback

$$K = t_1 \sum_{k=0}^{n-1} r_k A^k - t_1 \sum_{k=0}^{n-1} a_k A^k$$

$$= \left(t_1 \sum_{k=0}^{n-1} r_k A^k + t_1 A^n \right) - \left(t_1 \sum_{k=0}^{n-1} a_k A^k + t_1 A^n \right)$$

$$= t_1 \left(\sum_{k=0}^{n-1} r_k A^k + A^n \right) - t_1 \left(\sum_{k=0}^{n-1} a_k A^k + A^n \right)$$

This term is 0 by Cayley-Hamilton Theorem

$$= t_1 \left(\sum_{k=0}^{n-1} r_k A^k + A^n \right)$$

$$\therefore \boxed{K = \begin{bmatrix} 0 & 0 & \dots & 0 & 1 \end{bmatrix} \mathscr{C}^{-1} R(A)}$$

This is called Ackermann's Formula

- $R(s) = r_0 + r_1 s + \ldots + r_{n-1} s^{n-1} + s^n$ is the desired/target characteristic polynomial.

Example – Applying Ackermann's formula

- Suppose that

$$A = \begin{bmatrix} 0.514 & -0.057 & -0.972 \\ 0.076 & -0.238 & 0.152 \\ -0.243 & -0.0285 & 0.514 \end{bmatrix} \quad B = \begin{bmatrix} -0.1 \\ 1 \\ -0.2 \end{bmatrix}$$

  A has eigenvalues 1, -0.01, -0.2 with one eigenvalue in RHP.
- We can check that system is controllable.
- We want to move eigenvalues to -1, -2, -2 corresponding to $R(s) = (s+1)(s+2)(s+2) = s^3 + 5s^2 + 8s + 4$ by linear state feedback.

Example – Applying Ackermann's formula

- Then by Ackermann's Formula the required gain K in $u(t) = -Kx(t)$ is

$$K = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \mathscr{C}^{-1} R(A) = \begin{bmatrix} 66.6867 & -0.4706 & -64.6463 \end{bmatrix}$$